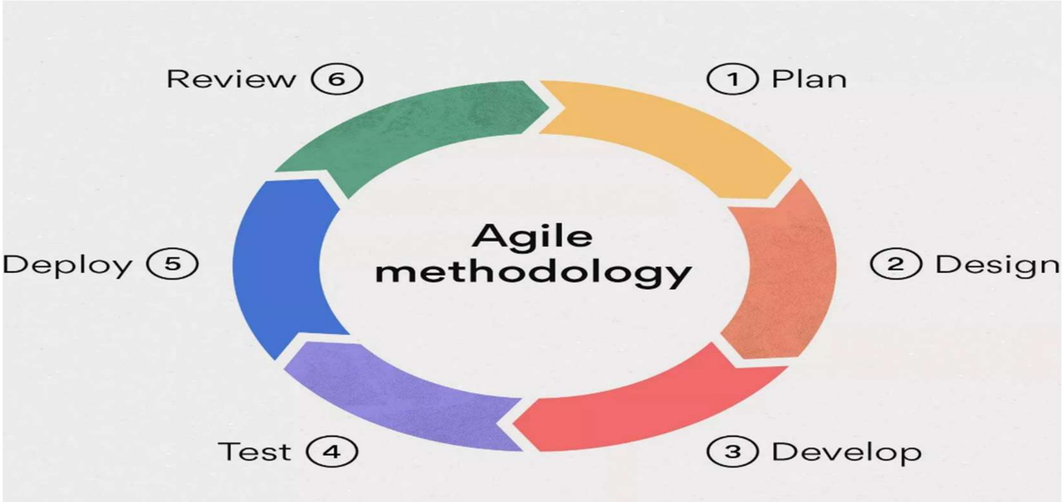
Methodology Adopted:

The project follows the Agile development methodology, allowing for iterative design, development, and testing. This approach ensures that the system can evolve based on user feedback and changing requirements. Regular sprints will be conducted, focusing on delivering functional increments of the system, with continuous integration and testing to maintain quality and address any issues promptly.



1. Project Initiation and Planning

- **Identify Stakeholders:** Determine who will be involved in the project, including developers, designers, testers, and end-users.
- **Define Project Scope:** Clearly outline the project's goals, features, and functionalities.
- **Create a Product Backlog:** Prioritize the features and functionalities into a backlog, considering user needs and business objectives.
- **Establish a Development Team:** Assemble a cross-functional team with the necessary skills to develop and maintain the e-library.
- **Set Up a Development Environment:** Configure the development environment, including software tools, servers, and databases.

2. Iterative Development

Sprint 1: Core Functionality

- **Sprint Planning:**
  - Select user stories from the product backlog.
  - Estimate the effort required for each user story.
  - Create a sprint backlog.

- **Development:**
  - Design and develop the core functionalities, such as user registration, login, book browsing, and reading.
  - Implement basic search and filtering capabilities.
- **Testing:**
  - Conduct unit and integration testing to ensure code quality.

STSYEM DESIGN

DFD/UML Diagram:

This context-level DFD for an e-library shows a simplified view of the system:

- **External Entities:**
  - **User:** A person who interacts with the system to read books.
- **Processes:**
  - **Read Book:** The core process where a user accesses and reads a book.
- **Data Stores:**
  - **Book Database:** Stores information about books, including title, author, genre, publication year, and content.

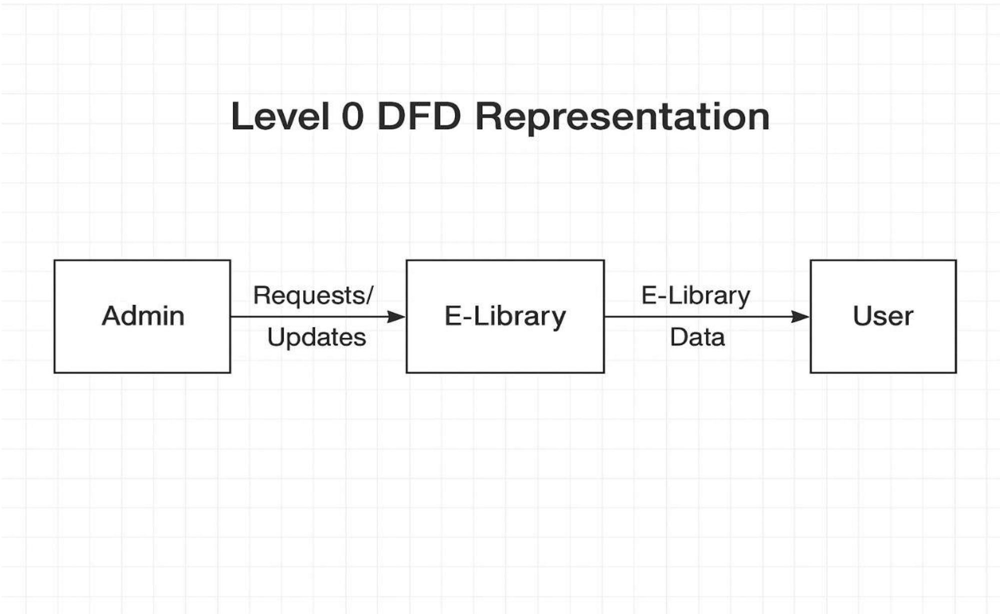


Figure 1: Diagram Represents Level 0 Data Flow Diagram

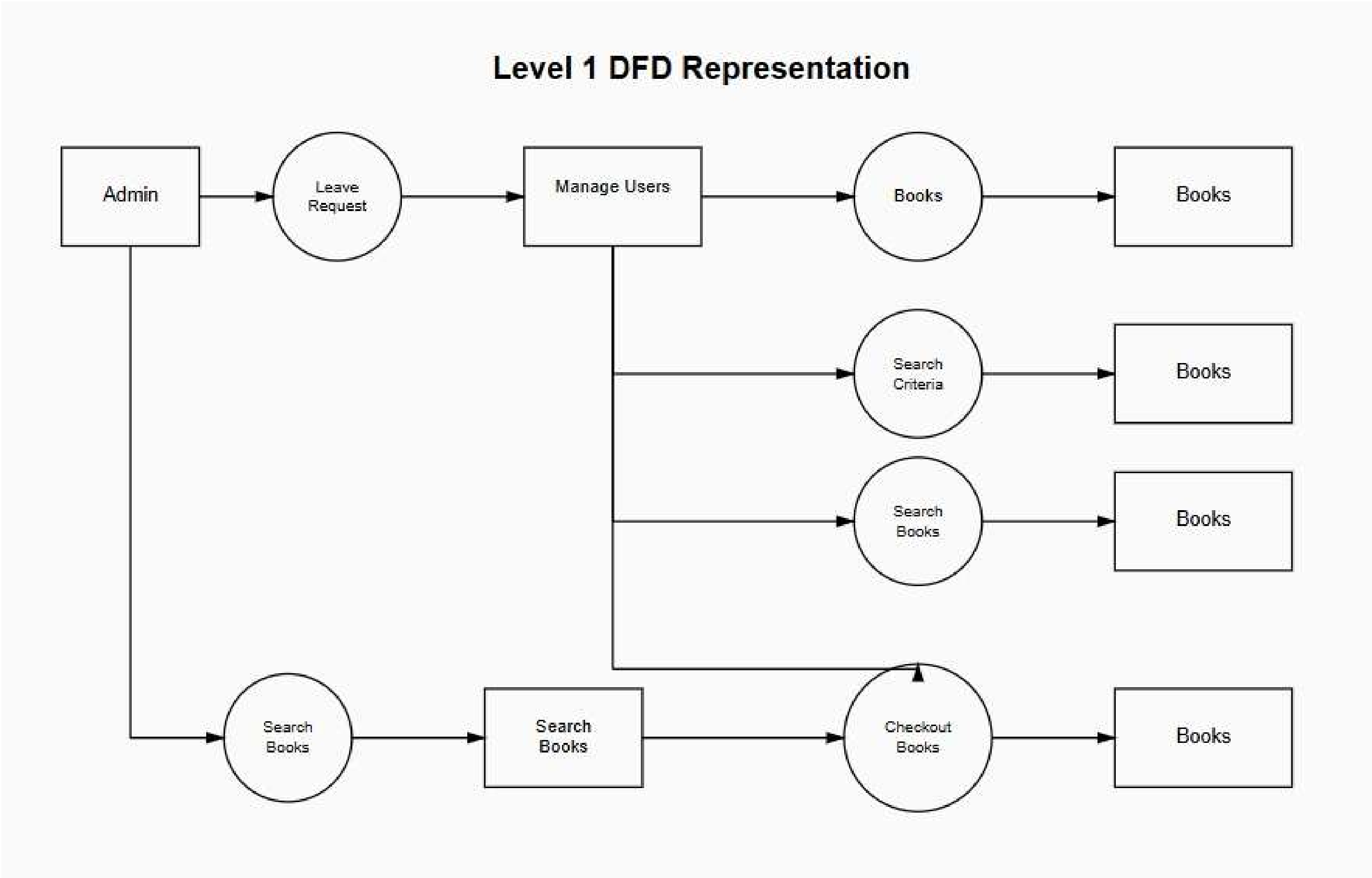


Figure 2: Diagram Represents Level 1 Data Flow Diagram

Entity Relationship Diagram: